

Auteur: Ibrahim Akiyev
Studierichting: Informatica
Oefeningenreeks 2D nr. 12.9

$$\begin{aligned} & \lim_{x \rightarrow \infty} (\sqrt{2x^2 + 3} - x) \\ & \lim_{x \rightarrow +\infty} \left(\frac{(\sqrt{2x^2 + 3} - x)(\sqrt{2x^2 + 3} + x)}{(\sqrt{2x^2 + 3} - x)} \right) \\ & \lim_{x \rightarrow +\infty} \frac{2x^2 + 3 - x^2}{(\sqrt{2x^2 + 3} - x)} \\ & \lim_{x \rightarrow +\infty} \frac{x^2 + 3}{(\sqrt{2x^2 + 3} - x)} \\ & \lim_{x \rightarrow +\infty} \frac{x}{(\sqrt{2 + 1})x} = +\infty \\ & \lim_{x \rightarrow -\infty} (\sqrt{2x^2 + 3} - x) = +\infty - (-\infty) = +\infty \end{aligned}$$

References